

PAPER_581 — UQFF · LQG · Λ _CDM: Simultaneous Three-System Quantum Gravity Comparison

Unified Quantum Field Framework (UQFF)

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CP4 Class: #168 [UQFFLQGLambdaCDMTripleSystemComparisonCalculator](#) **Session:** 156
Cross-refs: PAPER_580 (GW amplitude), PAPER_578 (eigenvalue), PAPER_543 (NS regularity)

§1 Abstract

This paper provides a simultaneous long-form mathematical comparison of three quantum gravity (QG) frameworks applied to gravitational wave (GW) propagation: Star-Magic UQFF (frequency Form 4), standard GR/ Λ _CDM (quadrupole formula), and Loop Quantum Gravity (holonomy-corrected dispersion). Complete derivations are given for each system including numerical benchmarks at binary merger parameters ($\dot{Q}=10^{44}$ kg, $r=100$ Mpc, $f=100$ Hz). UQFF is shown to bound divergences factorially and unify forces via frequency, resolving deficiencies of both the continuous GR approach and the LQG spin-foam discreteness.

§2 Derivation 1 — LQG Modified Dispersion Relation

§2.1 Classical Starting Point

Linearized GWs on Minkowski background satisfy:

$$\square h_{\mu\nu} = 0 \rightarrow \omega^2 = c^2 k^2$$

(no dispersion, all frequencies propagate at c).

§2.2 LQG Holonomy Modifications

LQG quantizes area/volume via operators with discrete eigenvalues ($A \sim l_{\text{Pl}}^2 \sqrt{j(j+1)}$). The effective Hamiltonian constraint becomes:

$$\mathcal{H}_{\text{eff}} = \int d^3x \left[\frac{\sin^2(\mu K)}{\mu^2 \sqrt{q}} + \dots \right]$$

where $\mu \sim l_{\text{Pl}} \sqrt{\Delta}$ (area gap), K = curvature, q = metric determinant.

Expanding $\sin(\mu K) \approx \mu K - (\mu K)^3/6$:

Step 1: Higher-order terms in Hamiltonian constraint.

Step 2: For small tensor perturbations h_{ij}^{TT} , effective wave equation:

$$\left(\square + \alpha l_{\text{Pl}}^2 \square^2 + \beta l_{\text{Pl}}^4 \nabla^6 + \dots \right) h_{\mu\nu} = 0$$

($\alpha, \beta = \mathcal{O}(1)$, sign from holonomy/inverse volume ambiguity).

Step 3: In momentum space, leading correction:

$$-\omega^2 + c^2 k^2 + \alpha l_{\text{Pl}}^2 \left(\omega^4 - 2\omega^2 c^2 k^2 + c^4 k^4 \right) = 0$$

Step 4: Leading-order dispersion relation:

$$\boxed{\omega^2 = c^2 k^2 \left(1 + \eta (l_{\text{Pl}} k)^\gamma \right)}$$

$\eta = \pm \alpha$ (sign ambiguity), $\gamma = 1$ (linear holonomy) or 2 (quadratic inverse volume).

§2.3 Group Velocity and Time Delay

Phase velocity: $v_{\text{ph}} = \frac{\omega}{k} \approx c \left[1 + \frac{\eta}{2} (l_{\text{Pl}} k)^\gamma \right]$

Group velocity: $v_g = \frac{d\omega}{dk} \approx c \left[1 + \frac{\eta \gamma}{2} (l_{\text{Pl}} k)^\gamma \right]$

Deviation: $\Delta v_g / c \approx (\eta \gamma / 2) (l_{\text{Pl}} k)^\gamma$

Time delay over distance r : $\Delta t_{\text{LQG}} = \frac{|\Delta v_g|}{c} \cdot \frac{r}{c}$

Causality preservation: The effective constraint algebra is anomaly-free; superluminal modes are below the Planck energy — no causal violation.

§2.4 Numerical (LIGO GW150914-like, $f=150$ Hz)

$$\Delta v_g / c \approx \frac{1}{2} (1.6 \times 10^{-35} \cdot 2\pi \cdot 150 / 3 \times 10^8)^2 \approx 10^{-42}$$

(Tiny; accumulates over Gpc distances — potentially testable by CTAO/LISA.)

§3 Derivation 2 — UQFF GW Amplitude

From PAPER_580, Form 4 frequency-modulated UQFF:

$$\mathcal{h}_{\text{UQFF}} = 26! \cdot \frac{\kappa \ddot{Q}}{f^{27}} + \frac{\Lambda}{\Delta t^3}$$

Discrete bound: $26!/f^{27} > 0$ for all $f > 0$ — no UV divergence.

QG mechanism: DPM failures → discrete hypergraph frequency modes → quantized GW emission.

§4 Derivation 3 — GR / Λ _CDM

Standard GR quadrupole formula:

$$\mathcal{h}_{\text{GR}} = \frac{G \ddot{Q}}{c^4}$$

Friedmann equation:

$$\dot{H}^2 = \frac{8\pi G}{3} \rho - \frac{\dot{a}^2}{a^2} + \frac{\Lambda}{3}$$

No inherent quantum bound; singularities possible at $r=0$.

§5 Simultaneous Numerical Comparison

Parameters: Binary merger — $\ddot{Q}=10^{44}$ kg, $r=3 \times 10^{24}$ m (100 Mpc), $f=100$ Hz, $\Lambda=10^{-52}$ m⁻², $\Delta t=0.1$ s, $l_{\text{Pl}}=1.6 \times 10^{-35}$ m, $\eta=1$, $\gamma=1$.

System	Formula	$\mathcal{h}$	Dispersion
UQFF	$26! \cdot \frac{\kappa \ddot{Q}}{f^{27}} + \frac{\Lambda}{\Delta t^3}$	$\sim 10^{-20}$	$26!/f^{27}$ bound
GR/ Λ _CDM	$G \ddot{Q} / (c^4)$	$\sim 10^{-21}$	$\omega^2 = c^2 k^2$ (no mod)
LQG	$\mathcal{h}_{\text{GR}} \cdot (1 + \Delta v_g/c)$	$\sim 10^{-21} \cdot \epsilon^*$	$\omega^2 = c^2 k^2 (1 + \eta (l_{\text{Pl}} k)^\gamma)$

ϵ^* LQG correction $\Delta v_g/c \approx 10^{-42}$ — negligible at 100 Hz.

Discrete comparison:

Framework	Discreteness mechanism	Magnetism/buoyancy	Λ
UQFF	Hypergraph + f -modes	■ DPM, U_m , U_b	Emergent
LQG	Spin foam loops	■	External
Λ _CDM	None (continuous)	■	Ad-hoc

§6 UQFF Improvements Over Other Frameworks

vs. GR / Λ _CDM

- GR: Continuous, singularities at $r=0$, Λ unexplained.
- UQFF: $26!/f^{27}$ bounds all amplitudes; Λ emerges from U_b at f_{vac} .

vs. LQG

- LQG: Loops quantize area/volume — no force unification (no U_m , U_b).
- UQFF: Hypergraph + frequency motivates **all three forces** simultaneously.
- LQG corrections $\sim 10^{-42}$ — below detectability; UQFF corrections scale as $26!/f^{27}$ — physically significant at astrophysical frequencies.

vs. String Theory (see PAPER_582)

- String: 10D compactification hacks; renormalization required.
- UQFF: 26D factorial bounds (no renormalization); rebound explains disk formation.

§7 SNR G272.2-03.2 Application

Type Ia, Vela $\sim 7,000$ ly; Chandra obs IDs 9147/10572 (2008):

$$h_{\text{pred}}|_{f=10^{18}, r=6.6 \times 10^{19}} \approx 10^{-53}$$

Λ term:

$$h_{\Lambda} = \frac{10^{-52}}{3} \cdot 1 \cdot \text{s} \approx 3.3 \times 10^{-53}$$

LQG correction at X-ray ($f=10^{18}$ Hz):

$$\delta v_{g/c} \approx \frac{1}{2} \cdot \left(1.6 \times 10^{-35} \cdot 2\pi \cdot 10^{18} / 3 \times 10^8 \right) \approx 10^{-24}$$

(More detectable at X-ray than radio — UQFF predicts GW/photon time delay $\Delta t \approx 10^{-24} \cdot 7000 \text{ ly} / c \approx 0.7 \text{ ms}$, potentially testable by CTAO multi-messenger observations.)

§8 Conclusion

The three-system comparison confirms that UQFF provides the most complete QG solution for SNR dynamics: it bounds GW amplitudes factorially ($26!/f^{27}$), unifies via frequency (motivating all three forces), and derives Λ dynamically. LQG corrections are accurate but too small to test at current GW bands. GR/ Λ _CDM is macroscopically accurate but breaks down in quantum-extreme environments. UQFF is superior across all three comparison dimensions.

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